Orca's Domains in MDVLCraft

What JujutsuCraft's domain expansions are, how its domain system works, which domain fits each archetype, and how hard it would be to bring them into the MDVLCraft Binder without JujutsuCraft as a dependency, paid for in mana. Research only: nothing here is implemented.

Source: JujutsuCraft ver51 (Forge 1.20.1, Modrinth), decompiled with Vineflower on 30 Oct 2026, with Orca's permission. The decompiled domain code is in `docs/orca-domains/src/` (146 domain classes and the procedures they call). Credit: Orca (JujutsuCraft); the binder README already credits him.

1. Summary

- **28 player domains** plus 7 NPC-only ones (Dabura, Inumaki, Kugisaki, Jinichi, Uro's object, the Angel and the "Object" props). Each has a number (`skill_domain`), a block palette for its inside, and an effect procedure run while it stands.
- **The domain core is separable**: casting, the barrier sphere, sure-hit, Simple Domain erosion, open barriers, clashing, breaking and restoring the world are about 3,500 lines in a dozen procedures, 18 barrier blocks and one mob effect.
- **The domains' effects are not**: most call JujutsuCraft's shared attack system (`RangeAttack`, `LogicAttack`, `LogicAttackDomain`) and its technique entities (Divine Dogs, Nue, Judgeman, Takada, the slot machine, the sea serpent...). Porting them as-is would drag in a large part of the mod.
- **Recommendation**: port the domain core and the barrier blocks and models; rewrite each domain's effect in the binder on top of what the pack already has (Iron's spells, Cursed Fate's entities, binder damage types). Pay for it with mana through the binder's existing domain upkeep (it already does this for the Malevolent Shrine).
- **Swapping the opening** (Orca's hand sign and barrier build, then Cursed Fate's shrine model and slashes inside): **feasible, medium effort**. JujutsuCraft's domain 1 already uses a bone-and-red palette made for the shrine.
- **Mana instead of cursed energy**: **easy**. Every domain's cost goes through one call (`ChangeCurseEnergyProcedure`); the binder replaces it with Iron's mana, as `ability/DomainUpkeep` already does.
- **No dependency**: **yes**, provided the effects are rewritten. Only the core, 18 blocks, 8 prop models and one player animation need copying.

2. How JujutsuCraft's domains work

Casting (the opening)

- The caster picks the domain as a technique (`select` = the domain number) and starts it. `DomainExpansionCreateBarrierProcedure` runs while the Domain Expansion effect is on them.
- **Hand sign**: the player plays `player_animation/domain_expansion1.json` (Player Animator, already in the pack) and is held in place (Slowness X) for 20 ticks while `cnt3` counts up. The cost (`used_technique_cost`) is taken at tick 0 through `ChangeCurseEnergyProcedure`.
- **Barrier**: from tick 20 the barrier grows outward over `cnt1` ticks to the radius (map variable `DomainExpansionRadius`, default 22; an open barrier uses 9 times that).

`DomainExpansionBattleProcedure` replaces blocks shell by shell: the wall with barrier blocks (tag `jujutsucraft:barrier`), the floor and fill with the domain's palette (`GetDomainBlockProcedure`). Each replaced block's id is kept in the new block's block entity (`old_block`) so it can be put back.

- When it completes, a crush sound plays and the domain's effect procedure starts (`DomainActiveProcedure` picks it by number).

While it stands

- Every 10 ticks `DomainExpansionOnEffectActiveTickProcedure` scans the sphere: a target under **Simple Domain** loses Simple Domain time (more against a stronger caster); everyone else gets the domain's **sure-hit** each second (`EffectCharactorProcedure`), and anyone without a domain of their own is **Neutralised** (can't use techniques) while inside.
- If the caster leaves the sphere, the domain ends. Damage to the caster is tracked (`oldHealth`) and lowers the domain's power.
- **Domain Amplification** (special key; costs cursed energy every tick and makes the user unstable) wraps the user in their own domain: the attack, anti-Infinity and sure-hit checks treat it as cancelling techniques on contact, which is how a target fights back inside an enemy domain.

Closing

- When the effect ends (`DomainExpansionEffectExpiresProcedure`), or the barrier is broken (`BreakDomainProcedure`), the barrier blocks turn back into the blocks they replaced, and the caster gets Brain Damage for 5 minutes (JujutsuCraft's domain cooldown: no new domain while it lasts).

3. Domain clashing

> **Power** = (Strength level + 10) × the caster's health fraction (current over maximum), × 1.15 for an open-barrier domain, × 1.5 for Deadly Sentencing, × 2 for Idle Death Gamble. When two domains overlap, each tick compares the two powers:
>
> - **One side ahead by 10 or more**: the weaker domain is *defeated* (`DomainDefeated`); its barrier gives way to the stronger one's and its sure-hit stops.
> - **Closer than that**: both are marked `Failed`; neither sure-hit lands and both barriers keep contesting the space (the Battle procedure keeps re-laying blocks) until one side's power drops (damage, effects running out) or a caster leaves.
> - **Open vs closed**: an open barrier (Sukuna and Kenjaku only, toggled with sneak) can't be shut in; players who have earned the "Countermeasures: Open Barrier" advancement reinforce their barrier against it. The Gojo, Higuruma, Yuta and Kenjaku NPCs count as already reinforced.
>
> For MDVLCraft the same rule maps cleanly to the binder: power from the caster's level or spell power instead of Strength, and the same 10-point margin. Cursed Fate's Malevolent Shrine doesn't take part in JujutsuCraft's clashes; it would need a shared binder registry of open domains to clash with ported ones (small: one map of active domains and one comparison per second).

4. Every domain

#	Domain	User	Inside (palette, props)	What it does (from the code)	Port
1	Malevolent Shrine	Sukuna	Bone floor, red sky	Endless Dismantle/Cleave slashes on everything (10.5+ damage each through `RangeAttack`) and blocks shredded in every direction; open-barrier capable.	**Medium**
2	Unlimited Void	Gojo	Universe blocks	Targets get Brain Damage, Slowness C, Blindness, Weakness C, techniques sealed and their cooldowns frozen: total paralysis while it holds.	**Easy**
4	Coffin of the Iron Mountain	Jogo	Stone and magma	Fire eruptions on everyone (flame explosion attacks).	**Easy**
5	Authentic Mutual Love	Yuta	Gravel field of katanas	Copies techniques into katanas and uses them; Jacob's Ladder circles; Rika.	**Hard**
6	Chimera Shadow Garden	Megumi	Shadow universe	Shadows slow and hold targets; summons Divine Dogs (black and white), Nue, Max Elephant and Toad from the shadows.	**Hard**
7	Kashimo's domain	Kashimo	Clouds	Lightning shocks on everything (electric attacks).	**Easy**
8	Horizon of the Captivating Skandha	Dagon	Beach: sky and sand	Sea serpents and fish shikigami swarm targets; caster gets Strength.	**Medium**
9	Tsukumo's domain	Yuki Tsukumo	Sand	Star Rage: the caster's virtual mass shoots up (huge hits); enemies slowed.	**Easy**
10	Choso's domain	Choso	Blood	Blood "water" entities and Piercing Blood ammo; Death Painting.	**Medium**
11	Mei Mei's domain	Mei Mei	Clouds and podzol	Crows flock and dive into targets.	**Easy**
12	Ishigori's domain	Ryu Ishigori	Universe, black sand	A giant cannon prop (animated model) fires Granite Blasts.	**Medium**
13	Nanami's domain	Nanami	Stone bricks	"Special": every target always has a 7:3 weak point (guaranteed critical hits).	**Easy**
14	Ceremonial Sea of Light	Hanami	Flowers under blue sky	Flowers drain targets: Weakness V and Slowness III.	**Easy**
15	Self-Embodiment of Perfection	Mahito	Bone hands	Idle Transfiguration on touch: 24-damage soul hits on everything.	**Easy**
18	Womb Profusion	Kenjaku	Blood and red	Gravity crush on everything inside (range attacks); open-barrier capable.	**Easy**
19	Time Cell Moon Palace	Uro	(entity prop)	Turns the sky over: 15-damage hits that strike from above.	**Easy**
20	Todo's domain	Todo	Flowers, blue sky	Boogie Woogie claps swap positions; Takada-chan appears; caster gets Strength.	**Medium**
21	Itadori's domain	Yuji	Clouds and podzol	Close-range soul strikes (range attacks).	**Easy**
23	Kurourushi's domain	Kurourushi	Bone, red	Cockroach swarms on every target.	**Medium**
24	Uraume's domain	Uraume	Ice	Frost attacks freeze everything.	**Easy**

25	Graveyard	(mod's)	Gravel, coffins, gravestones	Coffins and gravestones rise; seals techniques and Simple Domain; strength to the caster.	**Medium**
26	Ogi's domain	Ogi Zenin	Gravel	Flaming blade strikes (range attacks).	**Easy**
27	Deadly Sentencing	Higuruma	Courtroom planks	Judgeman holds a trial at a witness stand: guilty means Confiscation (techniques sealed) or the Executioner's Sword (instant death on second offence). 1,200 lines.	**Hard**
29	Idle Death Gamble	Hakari	White	A slot machine spins (animated prop); a jackpot gives near-immortality (Regeneration, Speed, Luck) for the jackpot song.	**Hard**
35	Junpei's domain	Junpei	(default)	Jellyfish (Moon Dregs) poison strikes.	**Easy**
36	Nishimiya's domain	Momo Nishimiya	(default)	Shikigami summoned; caster gets Strength.	**Medium**
40	Takuma Ino's domain	Takuma Ino	(default)	Kaichi (auspicious beast) summoned; caster gets Strength.	**Medium**
43	Rozetsu's domain	Rozetsu	(default)	Unstable cursed energy on targets; slows them.	**Easy**

NPC-only: Dabura's "Otherworld Between Darkness and Light: Reverse Transcendence" (its own model), Inumaki, Kugisaki, Jinichi (two models), Uro's object, Ishigori's object and the Angel (Jacob's Ladder). "Easy" = fits in the binder's existing toolkit (effects, binder damage, Iron's or Cursed Fate entities); "Medium" = needs a summon or prop entity ported; "Hard" = depends on large separate JujutsuCraft systems.

5. A domain for each archetype

Archetype	Domain	Why
Guardian	Tsukumo's domain (Star Rage)	Mass that turns the guardian into an immovable wall; enemies slowed around them.
Brute	Ishigori's domain	Raw output: a cannon that blasts everything, fitting the frontline bruiser.
Samurai	Ogi's domain	A swordsman's domain of blade strikes, without taking the Master Chef's shrine.
Berserker	Todo's domain	Boogie Woogie position swaps and a brawler's strength boost: charge in and keep switching targets.
Monk	Itadori's domain	Close-range soul strikes for an unarmed fighter.
Blaze	Coffin of the Iron Mountain	Fire eruptions everywhere.
Frost	Uraume's domain	Ice that freezes the whole sphere.
Water	Horizon of the Captivating Skandha	A beach and a sea of shikigami fish.
Lightning	Kashimo's domain	Lightning on everything inside.
Holy	Authentic Mutual Love	Rika, reverse cursed technique and the Jacob's Ladder light (healing and holy light).
Warlock	Unlimited Void	The void itself: total paralysis.
Necromancer	Graveyard	Coffins and gravestones rising; seals the enemy's arts. (Choso's blood domain is the alternative for blood arts.)
Druid	Ceremonial Sea of Light	A flower field that drains life. (Chimera Shadow Garden is the alternative for beasts.)

Jade	Time Cell Moon Palace	A celestial palace that turns the sky over.
Spirit	Self-Embodiment of Perfection	Soul manipulation, for the eastern spirit magic.
Assassin	Chimera Shadow Garden	Shadows to vanish into and shadow beasts that hunt the target.
Thief	Deadly Sentencing	Confiscation is the ultimate theft: the target's techniques are taken.
Stalker	Mei Mei's domain	Crows that scout and dive, for the ranged hunter.
Wanderer	Idle Death Gamble	The traveller's gamble: luck, speed and near-immortality on a jackpot.

Not used: Malevolent Shrine (the Master Chef and Shrinekin already have Cursed Fate's), Womb Profusion, Choso, Nanami, Kurourushi, Junpei, Nishimiya, Takuma Ino and Rozetsu. Nanami's guaranteed weak points would also suit the Samurai or Assassin if Ogi or Chimera prove too heavy.

6. Orca's opening with Cursed Fate's shrine and slashes

Feasible, medium effort. The two halves meet at a clear seam:

- **Opening (Orca)**: the hand-sign animation, the one-second hold, the barrier sphere built shell by shell from his barrier blocks, and the crush sound. Domain 1's palette (bone floor, red barrier) was made for the shrine.
- **Inside (Cursed Fate)**: the binder already drives Cursed Fate's Malevolent Shrine (`KitchenArts`, `DomainUpkeep`): its shrine model (`cursedfate:shrine`, open; the closed shrine entity), its sure-hit slashes (`SureHitEffectInDomainAreaProcedure`, already patched by the binder) and the binder's own slash damage (`ShrineSlashes`).
- **How**: when the shrine is cast, run the ported opening first (animation, hold, barrier build over about 2 seconds), then hand over to Cursed Fate's shrine and slashes exactly as now; on close, Orca's restore puts the world back. The open shrine (no barrier) would skip the build and use only the hand sign.
- **Watch for**: Cursed Fate's closed shrine may build its own enclosure; the binder would suppress that part with a mixin so the two don't overlap. Its radius (27) and Orca's (22) should be made the same.

7. Mana instead of cursed energy

Easy.

- JujutsuCraft takes a domain's cost in one place: `ChangeCurseEnergyProcedure.execute(entity, used_technique_cost)` at the start of the cast (and refunds it the same way). Its readiness check reads `PlayerCursePower`.
- In a binder port both become Iron's mana: check `MagicData.getMana()` before casting and take the cost with `SpecKit.addMana`.
- The binder already runs a domain on mana: `ability/DomainUpkeep` charges 10 mana a second for the Malevolent Shrine, closes it when the player can't pay, and keeps Cursed Fate's cursed energy topped up so Cursed Fate never stops it. Every ported domain would use the same upkeep (an opening cost, then mana per second), so domains drain mana like the shrine does now.

8. Bringing them in without JujutsuCraft

What to copy

- **Domain core** (about 3,500 lines of MCreator code to clean up into binder classes): create barrier, battle/build, active tick, sure-hit dispatch, simple domain and amplification, break, expiry/restore, clash power.
- **18 domain blocks** with their block entities (barrier, bone, blood, cloud, flower, ice, sand, universe and the rest), their models, textures and loot tables (they drop nothing).
- **Prop models** (GeckoLib): Ishigori's cannon, Uro's object, Rozetsu's, Dabura's, Inumaki's, Kugisaki's and Jinichi's two, with textures and animations.
- **Player animations**: `domain_expansion1`, `simple_domain1`/`2` (Player Animator is already in the pack).
- **Mob effects**: Domain Expansion, Simple Domain, Domain Amplification, Neutralisation (the binder can define its own with the same behaviour).

What to rewrite rather than copy

- Each domain's effect procedure. They call JujutsuCraft's attack system and its technique entities; copying them would mean copying most of the mod. Rewritten on binder tools they become short: most are a few effects and a damage pulse once a second (the "Easy" rows), some need a summon (Medium), and Deadly Sentencing, Idle Death Gamble, Authentic Mutual Love and Chimera Shadow Garden are their own projects (Hard).
- JujutsuCraft's map variables (`DomainExpansionRadius`) and its persistent-data counters become plain binder fields and a config value.

Effort

Piece	Estimate
Domain core, blocks, props, animations, mana upkeep, clash	2 sessions
The 19 archetype domains: 12 Easy, 4 Medium, 3 Hard (Holy, Assassin, Thief, Wanderer have the Hard ones)	Easy ½ session each; Medium 1; Hard 2+
Orca opening for Cursed Fate's shrine	1 session
Testing (two players are needed for clashes; this container fits one client)	1 session, plus a two-player check by you

Performance

- Not adding JujutsuCraft keeps its 6,500 files, 450 items and their registries out of the pack (load time and memory).
- The barrier build replaces roughly 6,000 blocks (a radius-22 shell) over about two seconds, each with a small block entity; restoring does the same once. Fine for a short burst; several domains at once on a busy server is the case to test.
- The per-10-tick sphere scans are cheap.

9. Risks

- MCreator code relies on string-keyed persistent data and implicit order between procedures; a straight port can break in subtle ways. Rewriting into binder classes is

safer than copying verbatim.

- Barrier blocks write into the world: a crash mid-domain leaves barrier blocks behind unless the restore also runs on server start (JujutsuCraft keeps the old block in each block entity, so a cleanup pass is possible).
- Clashing between ported domains and Cursed Fate's shrine needs the shared binder registry described in section 3.
- Orca is remaking the mod; this report reflects ver51 only.